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TEC–611

B. TECH. (ECE) (SIXTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

CMOS DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the following : (CO1)

(i) Etching

(ii) Dry and Wet Oxidation

(iii) CZ Process

OR

(b) Draw and explain NMOS fabrication process. (CO1)

P. T. O.

2. (a) Explain the following : (CO1)

(i) Photolithography

(ii) Diffusion

(iii) Metallization

OR

(b) Explain CMOS Latch-up problem. Explain p-well, n-well and twin well CMOS technology. (CO1)

3. (a) Explain CMOS stick diagram rules and draw stick diagram of CMOS NOR gate. (CO2)

OR

(b) Draw expression and stick diagram of expression : (CO2)

$$\overline{(A + B).(C + D + E)}$$

using CMOS technology.

4. (a) Draw expression and stick diagram of : (CO2)

$$\left[\overline{(A \,.\, B) + C + (D.E)}\right]$$

Gate using CMOS stick diagram rules.

OR

(b) Design stick diagram of NAND, NOR, EX-OR using CMOS technology. (CO2)

5. (a) Explain Lambda based design rules used for logic expressions layout design. (CO2)

OR

(b) Design the layout of Inverter using Lambda based design rules. (CO2)